Parametric Study of an Air Conditioning System

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Abstract

The report is a parametric study of an air conditioning system in summer conditions. This system is comprised of a mixing chamber, cooling coil, fan and conditioned space and is evaluated based on psychometric principles and energy balance equations. The analysis examines how the percentage of fresh air, the total cooling load, and sensible heat ratio (SHR) influence the performance of the system, in the context of supply air flow rate and cooling coil capacity. Findings suggest that, as the percentage of fresh air is increased, cooling coil load also increases proportionally because of the higher enthalpy of outdoor air and cooling load in turn causes the airflow and refrigeration load to be increased in the same manner. Also, the reduced SHR will cause higher latent cooling demand, which influences system performance. The results emphasize the trade-off between ensuring good indoor air quality and energy reduction. The findings are useful in the effective design and operation of HVAC systems in hot climates.

1. Introduction

Air conditioning systems are critical to the work of present-day buildings, as they provide thermal comfort, air quality in the building, and aid productivity. Air conditioning is necessary in hot climate areas like summer operation as it promotes health and safety in addition to comfort. The systems are meant to regulate some important parameters of the environment such as temperature, humidity, air movement and air purity.

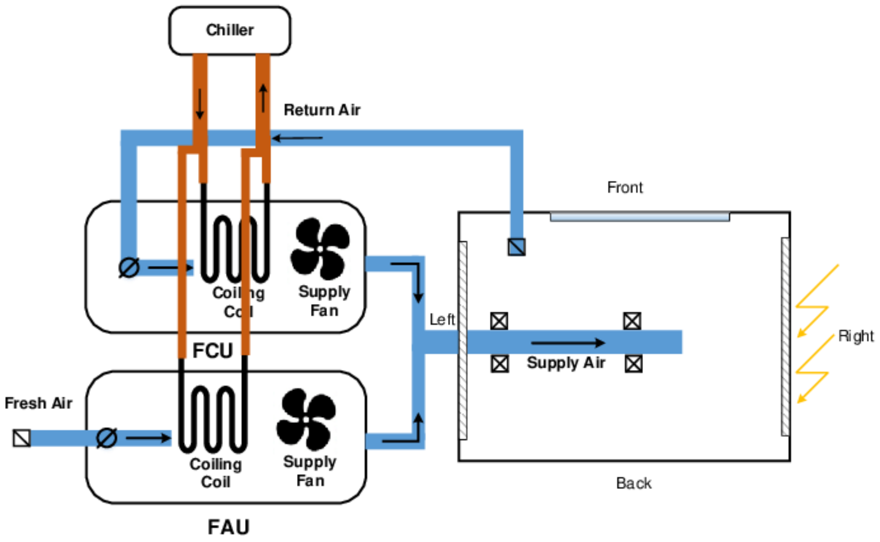

Figure 1:Typical HVAC system for a room

A combination of the relative humidity, mean radiant temperature, dry-bulb temperature, and air velocity determines thermo-comfort. Based on the set standard like those offered by ASHRAE, ensuring that the conditions indoors are maintained in a given range of comfort will mean that occupants do not experience a lot of thermal stress. In summer air conditioning the main task is to eliminate excess heat in indoor areas as well as regulating the moisture content by dehumidifying the air.

An average summer air conditioning system comprises of a few key elements such as a mixing chamber, cooling coil, fan and air distribution system. The mixing chamber is whereby the indoor air is recirculated and mixed with fresh outdoor air. The process is necessary since although recirculated air can save energy, fresh air is needed to ensure that the quality of indoor air is bearable, by diluting indoor air contaminants like carbon dioxide, odors and airborne pollutants.

One of the most important elements of the system is the cooling coil. It eliminates both sensible (related to decrease in temperature) and latent heat (related to the removal of moisture) of air. When warm and humid air flows over the cooling coil, its temperature drops and the moisture condensation occurs, producing conditioned air, which can be delivered to the inside area. The fan makes sure that the air circulates properly in the system, which overcomes the pressure losses, and provides the appropriate rate of airflow to achieve the needed conditions in the indoors.

Balancing indoor air quality and energy efficiency is one of the biggest problems to address when designing air conditioning systems. High-percentage of fresh outdoor air enhances air quality but also adds thermal loads particularly in hot climates where outdoor air is of high temperature and enthalpy. This causes more cooling needs and energy consumption by the cooling coil. Thus, engineers should cautiously consider the trade-offs of comfort, air quality, and energy consumption.

The other concept which is important in the analysis of air conditioning is the sensible heat ratio (SHR), which is the ratio of sensible heat load to total heat load. The SHR gives an idea of the significance of the control of temperatures as compared to the ones of humidity in a particular space. When SHR is lower, it implies a larger latent load, which means that the system needs more dehumidification, and hence more cooling.

The project is aimed at conducting a parametric study of a system of air conditioning that works under the conditions of summer. The paper examines the impact of crucial parameters of the percentage of fresh air, overall cooling load, and sensible heat ratio on the system performance. The study will use psychometric principles and energy balance equations in order to establish the necessary air flow rate and cooling coil capacity in varying operating conditions.

These relations are crucial to the effective design and performance of HVAC systems. The findings of this research offer rich information on the effects of decisions in design on system performance, energy use, and indoor environmental quality. This type of analysis is especially significant in the situation of growing energy requirements and the necessity of environmentally friendly building construction.

1.1. Problem Statement

The objective of this project is to perform a parametric study of an air conditioning system. The study aims to determine the supply air conditions, air flow rate, and cooling coil capacity under varying operating conditions.

2. Methodology

The objective of this project is to perform a parametric study of an air conditioning system. The study targets to determine the supply air conditions for HVAC, air flow rate, and cooling coil capacity under varying load operating conditions.

2.1. System Description

The air conditioning system considered in this study consists of several key components, each playing a critical role in maintaining indoor comfort conditions.

2.1.1. Mixing Chamber

The mixing chamber combines fresh outdoor air with recirculated return air from the conditioned space. This process is essential to maintain indoor air quality while minimizing energy consumption. The proportion of fresh air significantly affects the thermal load on the system.

2.1.2. Cooling Coil

The cooling coil is responsible for removing both sensible heat and latent heat from the air. As the mixed air passes over the coil, its temperature decreases and moisture condenses, resulting in dehumidified and cooled air suitable for indoor use.

2.1.3. Fan

The fan circulates air through the system, ensuring continuous airflow from the mixing chamber to the conditioned space. It overcomes system resistance and maintains the required air flow rate.

2.1.4. Conditioned Space

This is the indoor environment where temperature and humidity are controlled to achieve thermal comfort. The air conditioners absorbs heat and moisture from the space before being partially recirculated.

The introduction of fresh air is crucial for maintaining indoor air quality by removing contaminants, odors, and carbon dioxide. However, increasing fresh air also increases the cooling load because outdoor air typically has higher temperature and enthalpy. Therefore, a balance must be maintained between air quality requirements and energy efficiency.

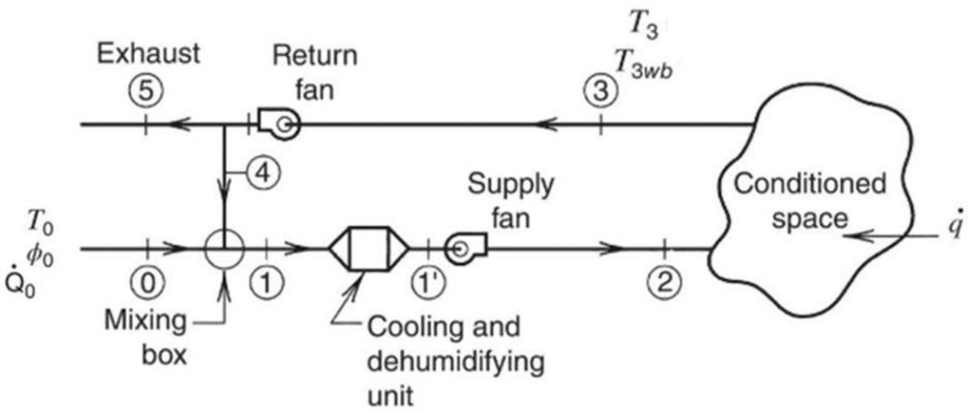

Figure 2:Air conditioning system

2.2. Assumptions

Indoor conditions are taken from ASHRAE[1]:

- Temperature = 25°C
- Relative Humidity = 50%
- Enthalpy = 50 kJ/kg

Outdoor conditions:

- Temperature = 40°C
- Relative Humidity = 40%
- Enthalpy = 85 kJ/kg

Base cooling load:

- Total load q = 10 kW
- Sensible load qs = 7 kW

Supply air:

- Temperature = 15°C
- Enthalpy = 30 kJ/kg

2.3. CALCULATIONS

To perform the analysis, the following conditions are assumed:

Indoor (Room) Conditions:

Indoor condition selected based on ASHRAE comfort zone

- Dry-bulb temperature: $T_r = 25°C$
- Relative humidity: $\phi_r = 50\%$

From psychrometric chart:

- Enthalpy: $h_r = 50\ kJ/kg$

Outdoor Conditions:

- Dry-bulb temperature: $T_o = 40°C$
- Relative humidity: $\phi_o = 40\%$

From chart:

- Enthalpy: $h_o = 85\ kJ/kg$

Base Cooling Load:

- Total load: $q = 10\ kW$
- Sensible load: $q_s = 7\ kW$

Supply Air Condition:

- Supply temperature: $T_s = 15°C$
- Enthalpy: $h_s = 30\ kJ/kg$

Energy balance equations are given by

Energy balance equation:

$$\dot{m} = \frac{q}{(h_{room} - h_{supply})}$$

Cooling coil load[2]:

$$q_{coil} = \dot{m}(h_{mixed} - h_{supply})$$

2.3.1. Base Case Calculation (30% Fresh Air)

Mixed Air Enthalpy is calculated by

$$h_m = (X \cdot h_o) + ((1 - X) \cdot h_r)$$
$$h_m = (0.3 \times 85) + (0.7 \times 50)$$
$$h_m = 60.5\ kJ/kg$$

Air Flow Rate is calculated using energy balance:

$$\dot{m} = \frac{q}{h_r - h_s}$$

$$\dot{m} = \frac{10}{50 - 30} = 0.5\ kg/s$$

Cooling Coil Capacity is calculated by

$$q_{coil} = \dot{m}(h_m - h_s)$$
$$q_{coil} = 0.5(60.5 - 30)$$
$$q_{coil} = 15.25\ kW$$

2.4. Case 1: Effect of Fresh Air Percentage

Air flow rate remains constant:

$$\dot{m} = 0.5\ kg/s$$

0% Fresh Air intake i.e full recirculation mode

$$h_m = 50$$
$$q_{coil} = 0.5(50 - 30) = 10\ kW$$

25% Fresh Air intake

$$h_m = (0.25 \times 85) + (0.75 \times 50) = 58.75$$
$$q_{coil} = 0.5(58.75 - 30) = 14.38\ kW$$

50% Fresh Air intake

$$h_m = 67.5$$
$$q_{coil} = 0.5(67.5 - 30) = 18.75\ kW$$

75% Fresh Air intake

$$h_m = 76.25$$
$$q_{coil} = 0.5(76.25 - 30) = 23.13\ kW$$

100% Fresh Air intake

$$h_m = 85$$
$$q_{coil} = 0.5(85 - 30) = 27.5\ kW$$

As fresh air increases, mixed air enthalpy increases → cooling coil load increases significantly.

2.5. *Case 2: Effect of Cooling Load*

For Case A cooling load $q = 15\ kW$ was assumed

$$\dot{m} = \frac{15}{50 - 30} = 0.75\ kg/s$$
$$q_{coil} = 0.75(60.5 - 30) = 22.9\ kW$$

For Case B: $q = 20\ kW$ was assumed

$$\dot{m} = \frac{20}{50 - 30} = 1.0\ kg/s$$
$$q_{coil} = 1.0(60.5 - 30) = 30.5\ kW$$

- Increasing cooling load increases airflow
- Cooling coil capacity increases proportionally

2.6. *Case 3: Effect of Sensible Heat Ratio (SHR)*

$$SHR = \frac{q_s}{q}$$

For Case A: SHR = 1.0 was assumed

$$q_s = 10\ kW$$

$$h_s = 32$$

$$\dot{m} = \frac{10}{50 - 32} = 0.56\ kg/s$$

$$q_{coil} = 0.56(60.5 - 32) = 15.96\ kW$$

Case B: SHR = 0.8 was assumed

$$q_s = 8\ kW$$

$$h_s = 30$$

$$\dot{m} = 0.5\ kg/s$$

$$q_{coil} = 15.25\ kW$$

For Case C: SHR = 0.6 was assumed

$$q_s = 6\ kW$$

$$h_s = 29$$

$$\dot{m} = \frac{10}{50 - 29} = 0.476\ kg/s$$

$$q_{coil} = 0.476(60.5 - 29) = 14.98\ kW$$

For Case D: SHR = 0.5 was assumed

$$q_s = 5\ kW$$

$$h_s = 28$$

- $\dot{m} = \frac{10}{50-28} = 0.455\ kg/s$

  $$q_{coil} = 0.455(60.5 - 28) = 14.79\ kW$$

  Lower SHR means more latent cooling
- Requires lower supply temperature
- Affects coil performance and humidity control

The air conditioning process was plotted on a psychrometric chart using PSYCH software. The outdoor air state (O) and return air state (R) were first located based on given temperature and relative humidity values. A straight line joining these two points represents the mixing process, and the mixed air state (M) was determined according to the percentage of fresh air. From the mixed air condition, a cooling and dehumidification process was drawn to reach the supply air condition (S). The chart was used to determine key properties such as enthalpy and humidity ratio for all state points[3].

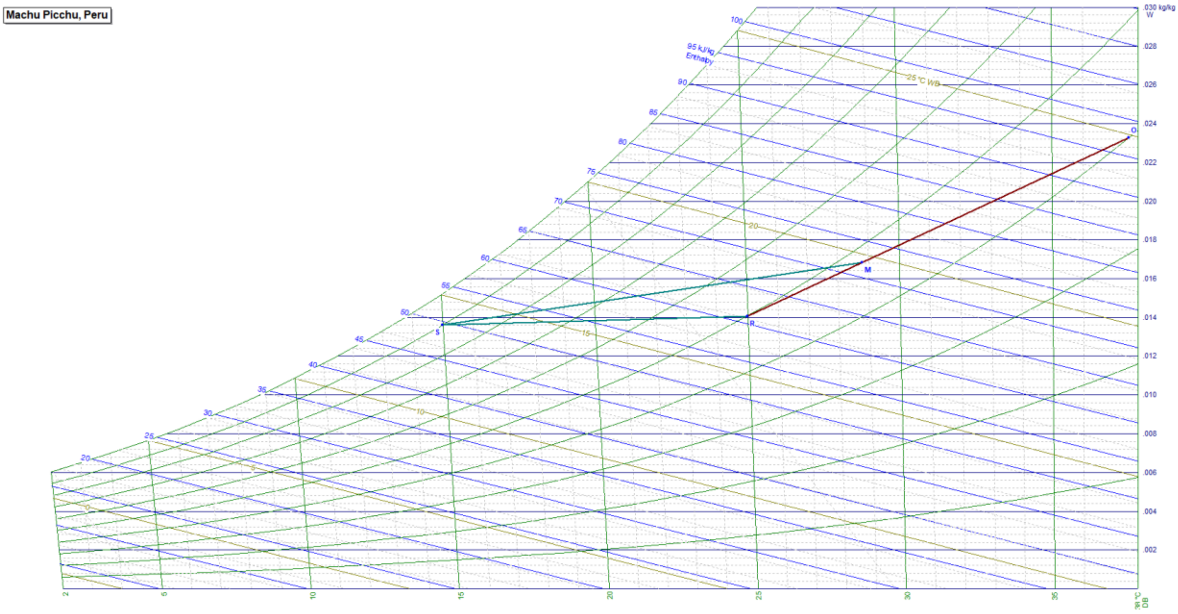

Figure 3:Psychometric Chart for 70% recirculation

3. Results and Discussion

The parametric analysis conducted in the research offers a clear insight into the effects of the key variables in a system on the operation of an air conditioning system in summer conditions. The results highlight the strong dependency of cooling coil capacity and airflow requirements on fresh air percentage, total cooling load, and sensible heat ratio (SHR).

3.1. Effect of Fresh Air Percentage

Among the most important findings made based on the results is the effect of raising the percentage of fresh air to the performance of the system. The enthalpy of the mixed air also increases with the increase in the proportion of the outdoor air because the outdoor air has a higher temperature and moisture content. This causes an enormous rise in cooling load being carried by the coil.

Table 1 :Effect of fresh air on cooling coil capacity

Fresh Air (%)	Mixed Air Enthalpy (kJ/kg)	Cooling Coil Capacity (kW)
0	50	10
25	58.75	14.38
50	67.5	18.75
75	76.25	23.13
100	85	27.5

It is evident that increasing fresh air from 0% to 100% results in nearly three times increase in cooling coil capacity.

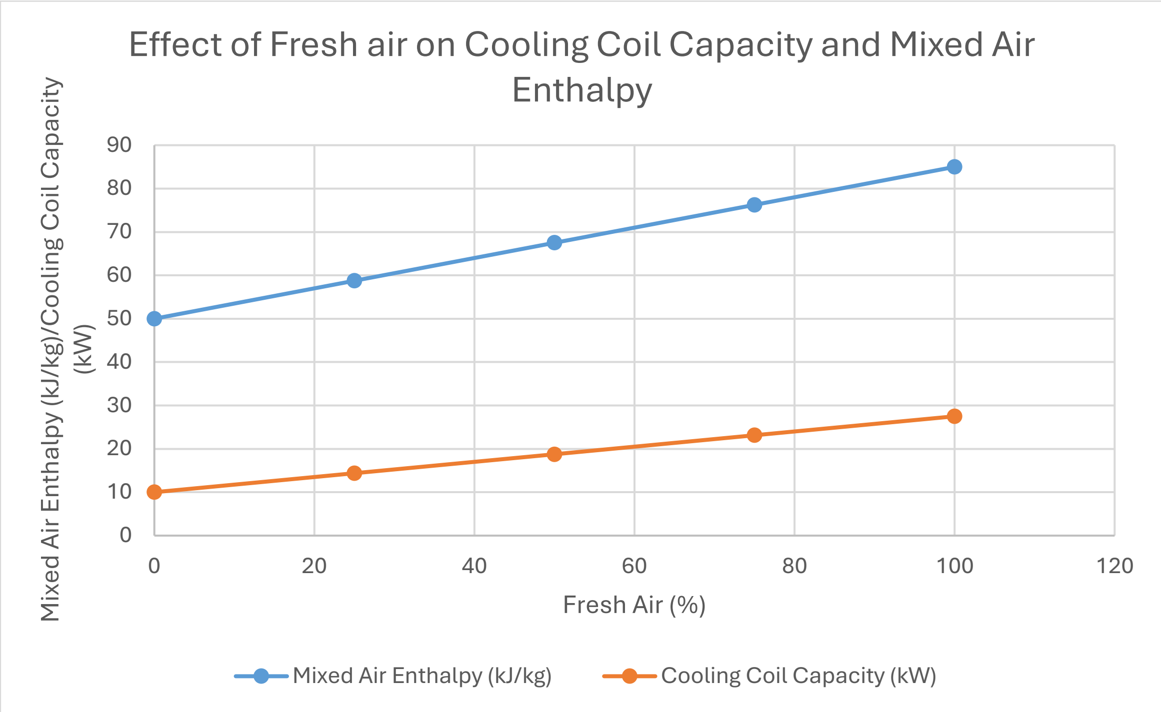

Figure 4:Effect of Fresh air on Cooling Coil Capacity and Mixed Air Enthalpy

This shows that although fresh air is needed to ensure a good Indoor air quality (as prescribed by ventilation guidelines like ASHRAE 62.1) it is also a major contributor to energy consumption. Thus, a balance between the needs in ventilation and the energy efficiency should be achieved.

3.2. *Effect of Cooling Load*

In the study it is also revealed that the cooling load directly and proportionally affects the airflow rate and cooling coil capacity. The more the total cooling load, the more conditioned air has to be supplied by the system, which has to take away the extra heat in the space.

Table 2:Effect of cooling load on cooling coil capacity

Total Load (kW)	Air Flow Rate (kg/s)	Cooling Coil Capacity (kW)
10	0.5	15.25
15	0.75	22.9
20	1.0	30.5

This linear relationship is a confirmation that the various components in a system like fans and cooling coils should be suitably sized to meet the peak loads effectively.

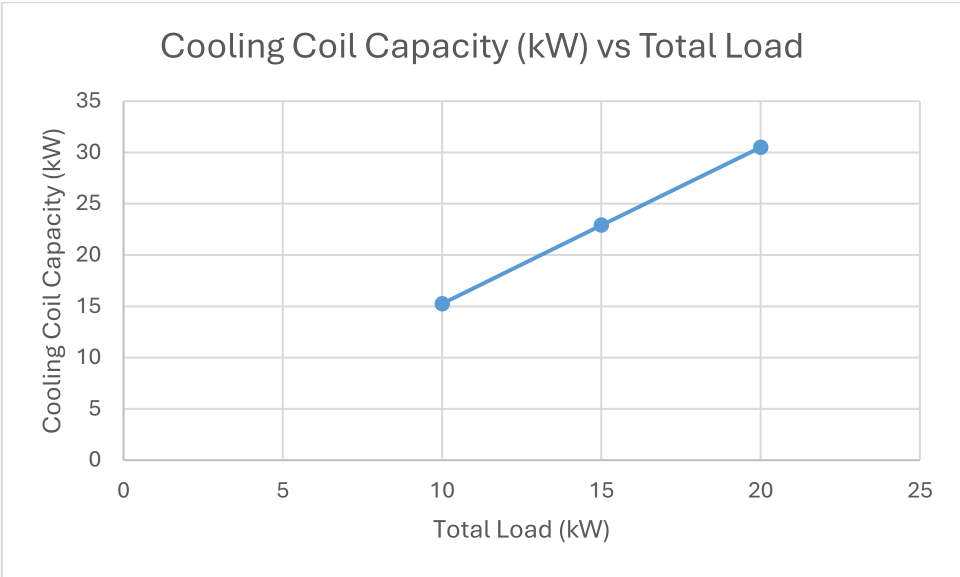

Figure 5:Effect of cooling load on cooling coil capacity

Excessive oversizing causes wastage of energy and inadequate undersizing causes poor comfort conditions.

3.3. Effect of Sensible Heat Ratio (SHR)

The sensible heat ratio is significant in the balance between the temerature control and the humidity control. Decreased SHR values imply increased latent heat loads, and demand increased dehumidification.

Table 3:Effect of Sensible Heat Ratio (SHR) on cooling coil capacity

SHR	Sensible Load (kW)	Cooling Coil Capacity (kW)
1.0	10	14.25
0.8	8	15.25
0.6	6	15.75
0.5	5	16.25

The cooling coil capacity also rises a bit with a decrease in SHR with additional latent cooling requirements.

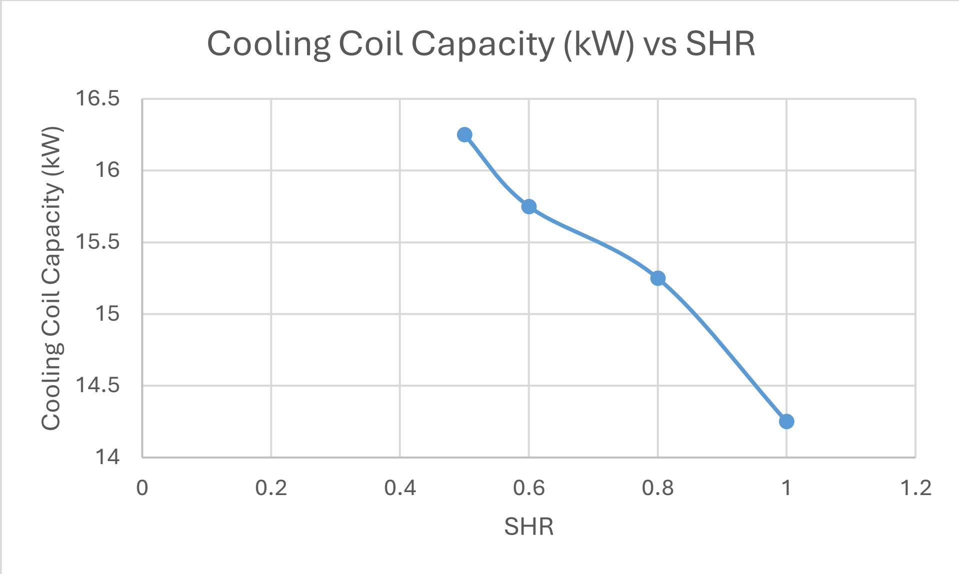

Figure 6:Effect of Sensible Heat Ratio (SHR) on cooling coil capacity

That is, the system will have to be run at lower supply air temperatures to get moisture out. Correct humidity level is needed to ensure comfort and avoidance of certain problems like mold development.

The results clearly demonstrate that:

- Fresh air will enhance the quality of the indoor air but will increase power usage considerably.
- Airflow and system capacity are highly affected by cooling load.
- Reduced SHR values enhance the latent cooling and dehumidification demands.

These results underline the significance of the meticulous system design. In choosing the operating conditions, engineers should take into account not only thermal comfort but also energy efficiency. The adoption of standards like ASHRAE will help to maintain acceptable indoor conditions, as well as reduce the unnecessary use of energy.

4. Conclusion

The current paper has been able to examine the performance of an air conditioning system under the operating conditions of summer in a parametric manner. The findings make it clear that the content of fresh air in the outdoors did play an important role in the performance of the system as more fresh air increased the mixed air enthalpy and resulted in a large increase in the cooling coil capacity. Although fresh air is necessary in ensuring the acceptable quality of air indoors and also ventilation is acceptable, it adds more thermal loads, thus increasing energy use. Moreover, analysis indicates that when total cooling load is increased, the rate of air flow as well as the capacity of the cooling coil is also increased in the same proportion, which demonstrates the significance of system sizing. The impact of sensible heat ratio indicates that, the lower the SHR values, indicating the higher the latent loads, the more dehumidification and a little more cooling power is needed. Overall, the study emphasizes the need to achieve an optimal balance between thermal comfort, indoor air quality, and energy efficiency. To design efficient and sustainable air conditioning systems, careful selection of operating parameters with the support of standard guidelines is needed.

5. REFERENCES

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[2]. Çengel, Y. A., & Boles, M. A., Thermodynamics: An Engineering Approach.

[3]. Psychrometric Analysis Software (PSYCH Tool).

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[5]. ASHRAE Standard 62.1 – Ventilation for Acceptable Indoor Air Quality.